SQL Interview Questions and Answers

1. What is SQL?

Answer:

- SQL stands for Structured Query Language. It is used to manage and communicate with databases.

2. What is a database?

Answer:

- A database is a collection of related data stored in an organized way so that it can be easily accessed, managed and updated.

3. What is a table in SQL?

Answer:

- A table is a collection of data organized in rows and columns. Each column has a name and each row is a record.

4. What is a primary key?

Answer:

- A primary key is a column that uniquely identifies each record in a table. It cannot have duplicate or NULL values.

5. What is a foreign key?

Answer:

- A foreign key is a column that refers to the primary key in another table. It helps to link two tables together.

6. What is the use of SELECT statement?

Answer:

- SELECT statement is used to retrieve data from one or more tables.

7. How do you select all columns from a table?

Answer:

- Use: SELECT * FROM table_name;

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7. How do you select all columns from a table?

Answer:

- Use: `SELECT * FROM table_name;`

Example:

```
SELECT * FROM students;
```

8. How do you select specific columns from a table?

Answer:

- Use: `SELECT column1, column2, ... FROM table_name;`

Example:

```
SELECT name, age FROM students;
```

9. What is the WHERE clause?

Answer:

- WHERE clause is used to filter records based on a condition.

Example:

```
SELECT * FROM students
WHERE age = 15;
```

10. What is ORDER BY clause?

Answer:

- ORDER BY clause is used to sort the result set in ascending (ASC) or descending (DESC) order.

Example:

```
SELECT * FROM students
ORDER BY name ASC;
```

11. What is ASC and DESC?

Answer:

- ASC → Ascending order (default).
- DESC → Descending order.

Example:

```
ORDER BY name ASC;   -- A to Z
ORDER BY name DESC;  -- Z to A
```

12. What is DISTINCT?

Answer:

- DISTINCT is used to remove duplicate values from the result.

Example:

```
SELECT DISTINCT city FROM students;
```

13. What is INSERT INTO?

Answer:

- INSERT INTO is used to add new records into a table.

Example:

```
INSERT INTO students (id, name, age)
VALUES (1, 'Rahul', 15);
```

14. What is UPDATE?

Answer:

- UPDATE is used to modify existing records in a table.

Example:

```
UPDATE students SET age = 16
WHERE id = 1;
```

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15. What is DELETE?

Answer:

- DELETE is used to remove one or more records from a table.

Example:

```
DELETE FROM students
WHERE id = 1;
```

16. What is NULL?

Answer:

- NULL represents empty or unknown value in a column.

Example:

```
INSERT INTO students (id, name, phone)
VALUES (2, 'Aman', NULL);
```

17. How do you check NULL values?

Answer:

- Use IS NULL to check for empty values.

Example:

```
SELECT * FROM students
WHERE phone IS NULL;
```

18. What is COUNT()?

Answer:

- COUNT() is an aggregate function that counts the number of rows.

Example:

```
SELECT COUNT(*) FROM students;
```

19. What is SUM()?

Answer:

- SUM() adds all the numeric values in a column.

Example:

```
SELECT SUM(marks) FROM students;
```

20. What is AVG()?

Answer:

- AVG() calculates the average of numeric values.

Example:

```
SELECT AVG(marks) FROM students;
```

21. What is MIN() and MAX()?

Answer:

- MIN() returns the smallest value and MAX() returns the largest value.

Example:

```
SELECT MIN(marks) FROM students;
SELECT MAX(marks) FROM students;
```

22. What is GROUP BY?

Answer:

- GROUP BY groups rows that have the same values in specified column(s).

Example:

```
SELECT class, COUNT(*)
FROM students
GROUP BY class;
```

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23. What is HAVING clause?

Answer:

- HAVING clause is used to filter records after GROUP BY.

Example:

```
SELECT class, COUNT(*)
FROM students
GROUP BY class
HAVING COUNT(*) > 5;
```

24. What is the LIKE operator?

Answer:

- LIKE is used to search for a specified pattern in a column.

Example:

```
SELECT * FROM students
WHERE name LIKE 'R%';
```

25. What do % and _ mean in LIKE?

Answer:

- % (percent) means any number of characters (including zero).
- _ (underscore) means any single character.

Example:

- `SELECT * FROM students WHERE name LIKE 'A%'; -- Names starting with A`
- `SELECT * FROM students WHERE name LIKE '_a%'; -- Second letter is a`

26. What is BETWEEN?

Answer:

- BETWEEN is used to select values within a range (inclusive).

Example:

```
SELECT * FROM students
WHERE age BETWEEN 13 AND 18;
```

27. What is IN operator?

Answer:

- IN is used to specify multiple values in a WHERE clause.

Example:

```
SELECT * FROM students
WHERE city IN ('Delhi', 'Mumbai', 'Kolkata');
```

28. What is JOIN?

Answer:

- JOIN is used to combine rows from two or more tables based on a related column.

Example:

```
SELECT s.name, c.class_name
FROM students s
JOIN classes c ON s.class_id = c.id;
```

29. What is INNER JOIN?

Answer:

- INNER JOIN returns only the matching records from both tables.

Example:

```
SELECT * FROM students s
INNER JOIN classes c
ON s.class_id = c.id;
```

30. What is LEFT JOIN?

Answer:

- LEFT JOIN returns all records from the left table and matching records from the right table.

Example:

```
SELECT * FROM students s
LEFT JOIN classes c
ON s.class_id = c.id;
```

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31. What is RIGHT JOIN?

Answer:

- RIGHT JOIN returns all records from the right table and matching records from the left table.

Example:

```
SELECT * FROM students s
RIGHT JOIN classes c
ON s.class_id = c.id;
```

32. What is FULL OUTER JOIN?

Answer:

- FULL OUTER JOIN returns all records when there is a match in either left or right table.

Example:

```
SELECT * FROM students s
FULL OUTER JOIN classes c
ON s.class_id = c.id;
```

33. What is CROSS JOIN?

Answer:

- CROSS JOIN returns the Cartesian product of both tables (every row of first table with every row of second table).

Example:

```
SELECT * FROM students
CROSS JOIN classes;
```

34. What is SELF JOIN?

Answer:

- SELF JOIN is a regular join, but the table is joined with itself.

Example:

```
SELECT a.name AS Employee, b.name AS Manager
FROM employees a
JOIN employees b
ON a.manager_id = b.id;
```

35. What is UNION?

Answer:

- UNION combines the result of two queries and removes duplicate rows.

Example:

```
SELECT name FROM students_2023
UNION
SELECT name FROM students_2024;
```

36. What is UNION ALL?

Answer:

- UNION ALL combines the result of two queries and keeps duplicate rows.

Example:

```
SELECT name FROM students_2023
UNION ALL
SELECT name FROM students_2024;
```

37. What is a SUBQUERY?

Answer:

- A subquery is a query written inside another query.

Example:

```
SELECT name FROM students
WHERE age > (SELECT AVG(age)
      FROM students);
```

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38. What is a CORRELATED SUBQUERY?

Answer:

- A correlated subquery depends on the outer query. It is executed once for each row processed by the outer query.

Example:

```
SELECT name FROM students s1
WHERE age > (SELECT AVG(age)
      FROM students s2
  WHERE s2.class_id = s1.class_id);
```

39. What is a NON-CORRELATED SUBQUERY?

Answer:

- A non-correlated subquery does not depend on the outer query. It is executed only once.

Example:

```
SELECT name FROM students
WHERE age > (SELECT AVG(age)
        FROM students);
```

40. What is a VIEW?

Answer:

- A view is a virtual table based on the result of a SELECT query. It does not store data physically.

Example:

```
CREATE VIEW student_view AS
SELECT id, name, age, class_id
FROM students;
```

41. What is an INDEX?

Answer:

- An index is a special data structure that improves the speed of data retrieval operations on a table.

Example:

```
CREATE INDEX idx_name
ON students(name);
```

42. What is DELETE, TRUNCATE and DROP?

Answer:

- DELETE – Removes specific rows.
- TRUNCATE – Removes all rows, but table structure remains.
- DROP – Deletes the table completely (structure + data).

Example:

```
DELETE FROM students WHERE id = 1;
TRUNCATE TABLE students;
DROP TABLE students;
```

43. What is PRIMARY KEY?

Answer:

- A primary key is a column (or set of columns) that uniquely identifies each record in a table. It cannot have duplicate or NULL values.

Example:

```
CREATE TABLE students (
  id INT PRIMARY KEY,
  name VARCHAR(50),
  age INT
);
```

44. What is FOREIGN KEY?

Answer:

- A foreign key is a column that refers to the primary key in another table. It helps maintain relationships between tables.

Example:

```
CREATE TABLE enrollments (
  id INT PRIMARY KEY,
  student_id INT,
  FOREIGN KEY (student_id) REFERENCES
     students(id)
);
```

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45. What is a TRANSACTION?

Answer:

- A transaction is a sequence of SQL operations that are treated as a single unit. Either all operations are successful, or none are.

Example:

```sql
BEGIN TRANSACTION;
UPDATE students SET age = 16 WHERE id = 1;
COMMIT;  -- saves the changes
```

46. What is ACID in DBMS?

Answer:

- ACID stands for:
  - A – Atomicity (All or nothing)
  - C – Consistency (Data remains correct)
  - I – Isolation (Transactions are independent)
  - D – Durability (Changes are permanent)

Example:

ACID properties ensure reliable processing of transactions in a database.

47. What is NORMALIZATION?

Answer:

- Normalization is the process of organizing data in tables to reduce redundancy and improve data integrity.

Example:

It helps to save space and avoid update, insert, and delete anomalies.

48. What is First Normal Form (1NF)?

Answer:

- 1NF says each column must have atomic (single) values and no repeating groups or arrays.

Example:

❌ Not 1NF:

ID	Name	Subjects
1	Rahul	Math, Science

✅ 1NF:

ID	Name	Subject
1	Rahul	Math
1	Rahul	Science

49. What is Second Normal Form (2NF)?

Answer:

- 2NF is in 1NF and all non-key attributes must depend on the whole primary key, not on part of it.

Example:

If a table has composite key (ID, Subject), then student name should depend on both ID and Subject, not only on ID.

50. What is Third Normal Form (3NF)?

Answer:

- 3NF is in 2NF and there should be no transitive dependency (non-key column should not depend on another non-key column).

Example:

If StudentID → ClassID and ClassID → ClassName, then ClassName should be in a separate table, not in the student table.

❌ Not 3NF

StudentID	StudentName	ClassID	ClassName
1	Rahul	10	10A

✅ 3NF

StudentID	StudentName	ClassID
1	Rahul	10

ClassID	ClassName
10	10A

BONUS SQL Interview Question & Answer

51. What is the difference between WHERE and HAVING?

Answer:

- WHERE filters rows BEFORE grouping.
- HAVING filters groups AFTER GROUP BY.

Example of WHERE

```sql
SELECT * FROM students
WHERE marks > 70;
```

→ First filters students whose marks are greater than 70.

Example of HAVING

```sql
SELECT class, COUNT(*)
FROM students
GROUP BY class
HAVING COUNT(*) > 5;
```

→ First groups students by class, then shows only classes having more than 5 students.

Easy Trick to Remember

Clause	Works On
WHERE	Individual rows
HAVING	Grouped data

Real-Life Example

- WHERE → Select students older than 15.

- HAVING → Select classes having more than 30 students.

Note:
Use WHERE when you want to filter rows.
Use HAVING when you want to filter groups.